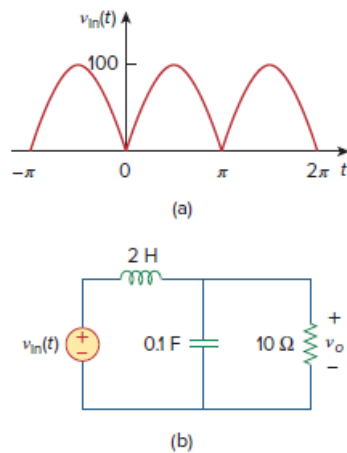


Problem

The full-wave rectified sinusoidal voltage in Fig. 17.77(a) is applied to the low-pass filter in Fig. 17.77(b). Obtain the output voltage $v_o(t)$ of the filter.

Figure 17.77:



Step-by-step solution

Step 1 of 5

Refer to Figure 17.77 (a) in the textbook.

Write the general Fourier series expression.

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Here,

$$\omega_0 = \frac{2\pi}{T} \text{ is the fundamental frequency.}$$

From the table 17.3(6) in the textbook, the Fourier series of the full wave rectified sinewave is

$$\begin{aligned} f(t) &= \frac{2A}{\pi} - \frac{4A}{\pi} \sum_{n=1}^{\infty} \left(\frac{1}{4n^2 - 1} \cos n\omega_0 t \right) \\ &= \frac{200}{\pi} - \frac{400}{\pi} \sum_{n=1}^{\infty} \left(\frac{1}{4n^2 - 1} \cos n\omega_0 t \right) \end{aligned}$$

Here,

Time period, $T = \pi$

Fundamental frequency is,

$$\begin{aligned} \omega_0 &= \frac{2\pi}{\pi} \\ &= 2 \end{aligned}$$

Therefore, the Fourier series of given full wave rectifier is,

$$v_{in}(t) = \frac{200}{\pi} - \frac{400}{\pi} \sum_{n=1}^{\infty} \left(\frac{1}{4n^2 - 1} \cos 2nt \right)$$

Step 2 of 5

Refer to Figure 17.77 (b) in the textbook.

Calculate the current through the source.

$$\begin{aligned} \mathbf{I}_s &= \frac{1}{(10) \left(\frac{10}{j\omega} \right)} \mathbf{V}_s \\ &= \frac{1}{j\omega 2 + \frac{10}{10 + \frac{10}{j\omega}}} \mathbf{V}_s \\ &= \frac{1}{j\omega 2 + \frac{10}{j\omega + 1}} \mathbf{V}_s \\ &= \frac{j\omega + 1}{j\omega 2 (j\omega + 1) + 10} \mathbf{V}_s \\ &= \frac{j\omega + 1}{j\omega 2 + 10 - 2\omega^2} \mathbf{V}_s \end{aligned}$$

Step 3 of 5

Calculate the current through $10 \, \Omega$ resistor using current division rule.

$$\begin{aligned} \mathbf{I}_o &= \frac{\left(\frac{10}{j\omega} \right)}{10 + \frac{10}{j\omega}} \mathbf{I}_s \\ &= \frac{1}{1 + j\omega} \mathbf{I}_s \\ &= \frac{1}{1 + j\omega} \cdot \frac{j\omega + 1}{j\omega 2 + 10 - 2\omega^2} \mathbf{V}_s \\ &= \frac{1}{j\omega 2 + 10 - 2\omega^2} \mathbf{V}_s \end{aligned}$$

Therefore, the output voltage is,

$$\begin{aligned} \mathbf{V}_o &= 10 \mathbf{I}_o \\ &= \frac{10}{j\omega 2 + 10 - 2\omega^2} \mathbf{V}_s \end{aligned}$$

Substitute $2n$ for ω_n .

$$\begin{aligned} \mathbf{V}_o &= \frac{10}{j4n + 10 - 8n^2} \mathbf{V}_s \\ &= \frac{5}{j2n + 5 - 4n^2} \mathbf{V}_s \end{aligned}$$

[Comments \(1\)](#)



Anonymous

why the substitution?

Step 4 of 5

For dc component, ($\omega_n = 0$ or $n = 0$),

$$\begin{aligned} V_{\text{dc}} &= \lim_{n \rightarrow 0} \left| \frac{10}{j4n + 10 - 8n^2} \times V_{\text{dc}} \right| \\ &= \lim_{n \rightarrow 0} \left| \frac{10}{j4n + 10 - 8n^2} \times \frac{200}{\pi} \right| \\ &= \frac{200}{\pi} \end{aligned}$$

Step 5 of 5

For the n-th harmonic,

$$\begin{aligned} \mathbf{V}_s &= -\frac{400}{(4n^2 - 1)\pi} \angle 0^\circ \\ &= \frac{400}{(4n^2 - 1)\pi} \angle 180^\circ \end{aligned}$$

Therefore,

$$\begin{aligned} \mathbf{V}_o &= \frac{5}{j2n + 5 - 4n^2} \frac{400}{(4n^2 - 1)\pi} \angle 180^\circ \\ &= \frac{5}{\sqrt{(5 - 4n^2)^2 + 4n^2} \angle \tan^{-1}\left(\frac{2n}{5 - 4n^2}\right)} \frac{400}{(4n^2 - 1)\pi} \angle 180^\circ \\ &= \frac{2000}{\pi(4n^2 - 1) \left[\sqrt{(5 - 4n^2)^2 + 4n^2} \right]} \angle \left[180^\circ - \tan^{-1}\left(\frac{2n}{5 - 4n^2}\right) \right] \end{aligned}$$

Therefore, the output voltage of the filter is,

$$\begin{aligned} v_o(t) &= \frac{200}{\pi} + \sum_{n=1}^{\infty} A_n \cos(2nt + \theta_n) \quad \text{V} \\ \text{Where} \\ A_n &= \frac{2000}{\pi(4n^2 - 1) \left[\sqrt{(5 - 4n^2)^2 + 4n^2} \right]} \\ \theta_n &= 180^\circ - \tan^{-1}\left(\frac{2n}{5 - 4n^2}\right) \end{aligned}$$